

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1: ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA04	Course Title:	Operating Systems
CO Assessed:	CO1 – Imitation (BL2, BL3)	Assessment:	Tool 2: Concept Mapping
Weightage:	20% of CIA (AT2 – 35%)	Total Marks:	20 (2 Questions × 10 Marks)
CO1 Statement:	Apply the concepts of Operating System types, Structures, and System calls to demonstrate their functions in various computing environments.		
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Slot / Dept:	Aim1	Date:	21-7-26

Instructions to Students

- Answer BOTH the questions. Each question carries 10 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.
- Original work is expected. Any form of plagiarism or copying from peers or online sources will be treated as academic misconduct.

Concept Mapping

10 Marks

Question 1

Concept Mapping: Operating System Components and Services

An Operating System provides several services to users and application programs.

Task:

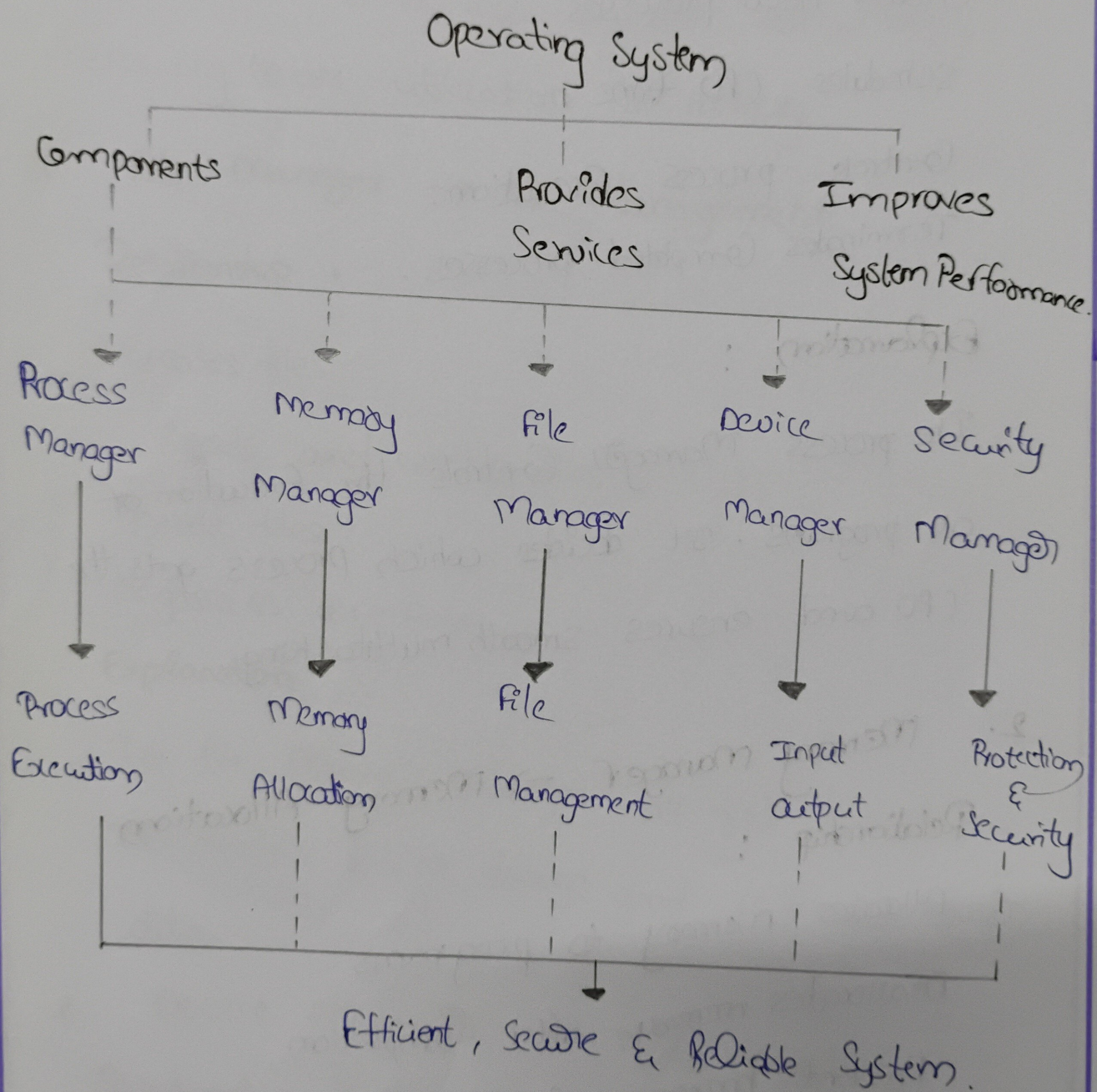
Match each Operating System Component with the most appropriate Operating System Service and explain the relationship.

Components	Services
Process Manager	File Management
Memory Manager	Process Execution
File Manager	Memory Allocation
Device Manager	I/O Operations
Security Manager	Protection & Security

After mapping, explain how these components collectively improve overall system performance.

Operating System Components & Services

Concept Map



1. Process Manager → Process Execution

Relationship :

Creates new processes

Schedules CPU time.

Controls process Execution.

Terminates Completed processes.

Explanation :

The process Manager Controls the Execution of all programs. It decides which process gets the CPU and ensures smooth multitasking.

2. Memory Manager → Memory Allocation

Relationship :

Allocates memory to programs.

Deallocates memory after Completion.

Prevents memory Conflicts.

Supports Virtual Memory.

Explanation :

The memory manager efficiently allocates RAM to processes, and free it after execution, ensuring better utilization of memory.

3. File Manager → File Management

Relationship :

Creates files

Reads and writes files

Delete files.

Organizes directories.

Explanation :

The file manager manages all files and folders. It provides secure storage and retrieval of user data.

4. Device Manager → I/O operations

Relationship :

Controls input/output devices.

Communicates with device drivers. ;
Manages data transfer.

Explanation :

The Device Manager Controls hardware devices like the keyboard, mouse, printer ... through I/O operations

5. Security manager → Protection & Security.

Relationship :

- User authentication,
- Access Control.
- File permission

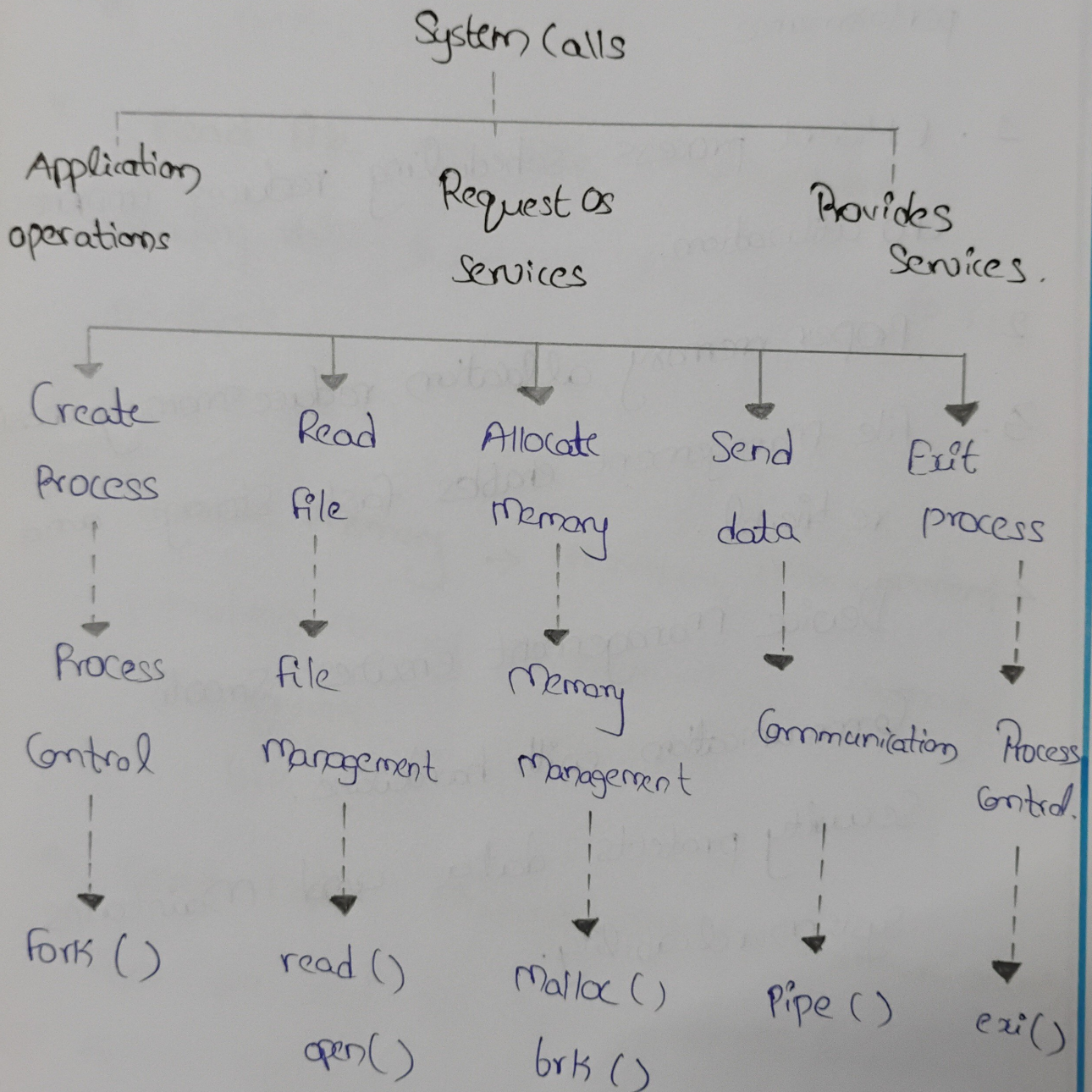
Protects Against unauthorized access.

Explanation :

The Security Manager ensures that only authorized users can access system resources and protects data from misuse.

System calls and Operating system services

Concept Map



Overall System Performance .

All Os Components work together to improve performance .

1. Efficient process scheduling reduces improves CPU utilization.
2. Proper memory allocation reduce memory wastage.
3. File Management enables fast Storage and retrieval.
4. Device Management ensures Smooth Communication with hardware.
5. Security protects data and Maintains System reliability.

1. Create process $\rightarrow$ Process Control.

Create a new process is controlled by the process control category.

Eg :- fork () .

2. Read file $\rightarrow$ File Management.

Reading data from a file belongs to file operations .

Eg :- read ()

3. Allocate memory $\rightarrow$ memory Management

Allocation memory is handled by memory Management .

Eg :- brk ()

4. Send Data through Pipe $\rightarrow$ Communication.

Pipes are used for inter - Process Communication.

Eg :- Pipe () .

5. Exit process $\rightarrow$ Process Control.

Terminating a process is a process control operating.

Eg :- `exit()`.

Conclusion

System calls provides an interface between user programs and the operating systems. They allow applications to perform tasks such as process creating, file access, memory allocation, communication and process termination. These services helps the OS manage system resources efficiently and securely.